Table 8.2 Common testing handlers in LangChain.js

Helper Class	Description
`FakeLLM`	Simulates a language model's behavior, allowing you to specify predefined responses or errors for testing purposes.
`FakeStreamingLLM`	Similar to `FakeLLM` but designed for scenarios where streaming responses are expected. It yields responses over time.
`FakeChatModel`	Simulates chat interactions by overriding the `_generate` method to return fixed messages based on input.
`FakeStreamingChatModel`	Extends `FakeChatModel` to support streaming responses, yielding messages over time while simulating chat behavior.
`FakeRetriever`	Simulates a retriever that returns a fixed set of documents when queried, useful for testing retrieval logic.
`FakeRunnable`	A fake runnable that can return input as output or options based on configuration, useful for testing runnable logic.
`FakeSplitIntoListParser`	Parses comma-separated values from text and returns them as an array, useful for testing parsing logic.
`FakeListChatModel`	A fake chat model that returns a predefined list of responses, useful for simulating sequential chat interactions.
`FakeEmbeddings`	Provides fake embeddings by overriding methods to return fixed values, useful for testing embedding-related functionality.
`FakeTracer`	A mock tracer that collects run data during tests, allowing you to verify tracing behavior without actual execution.
`FakeChatMessageHistory`	Simulates chat message history management, allowing you to test how your application handles message storage and retrieval.
`FakeListChatMessageHistory`	Similar to `FakeChatMessageHistory` but designed for list-based message management in chat applications.
`FakeTool`	A mock tool that can be used in testing scenarios to simulate tool behavior and validate interactions with tools.

Listing 8.9 Unit testing the RAG class methods using `FakeLLM`

```
describe('RAGSystem', () => {
  let ragSystem;

  beforeEach(() => {
    const apiKey = 'test_api_key';
    ragSystem = new RAGSystem(apiKey);

    const retriever = new FakeRetriever({
      output: [
        { pageContent: 'This is some context about AI.' },
        { pageContent: 'This is another context about machine learning.' },
      ],
    });
```

Initializes a new RAGSystem instance with a test API key for each test case

Creates a FakeRetriever instance with predefined output documents to simulate document retrieval